

Practice task for paraphrase

A study into child development published in 2010 was one of the first to demonstrate that childhood experience influences the structure of the developing brain. Since then, other studies have shown a link between a baby's socioeconomic status and the growth of its brain. Despite millennia of child rearing, we have only a limited understanding of how babies take such gigantic strides in cognitive, linguistic, reasoning and planning ability. At birth, the brain has nearly a hundred billion neurons, as many as in adulthood. As the baby grows, receiving a flood of input through their senses, neurons get connected to other neurons, resulting in some hundred trillion connections by the age of three.

Using new technology, scientists can better understand the mystery of how a child goes from being barely able to see when just born to being able to talk, ride a bike, draw, and invent an imaginary friend by the age of five. The more scientists find out about how children acquire the capacity for language, numbers and emotional understanding during this period, the more they realise that the baby brain is an incredible learning machine. Its future—to a great extent—is in our hands.

Judit Gervain, a **cognitive** neuroscientist at Paris Descartes University, **tested** how good newborns are at **distinguishing different sound patterns**. **Using near-infrared spectroscopy**, the researchers **produced images of the brains of babies** as they heard **audio sequences**. In some, the sounds were repeated in an **ABB structure**, such as mu-ba-ba; in others, an **ABC structure**, such as mu-ba-ge. The researchers found that **brain regions responsible for speech and audio processing** responded more strongly to the ABB sequences. In a later study, they found that the newborn brain was also able to distinguish between audio sequences with an AAB pattern and those with an ABB pattern. Not only could babies notice **repetition**, **they also were sensitive** to where it occurred in the sequence. Gervain is excited by these findings because the **order of sounds** is the building block of words and grammar. **'Position** is key to language,' she says. 'If something is at the beginning or at the end, it makes a big difference: "John killed the bear" is very different from "The bear killed John."'

Elsewhere, researchers led by Patricia Kuhl, a neuroscientist at the University of Washington in Seattle, have found that **language** delivered by **television, audio book, internet, or smartphone** — no matter how educational — doesn't appear to be enough for **children's development**. They carried out a study of nine-month-old American babies. The **researchers** expected the group who'd **watched videos** in Mandarin Chinese to show the same kind of **learning** as the group who were **face-to-face** with the same sounds. Instead they found a huge **difference**. The babies exposed to the language through **human interactions** were able to **distinguish** between similar Mandarin sounds as well as native listeners. But the other babies — regardless of whether they had watched the video or listened to the **audio** — showed no learning whatsoever.